BUSINESS REQUIREMENTS DOCUMENT

App Chấm Công — Time

Field	Value
Project Name	App Chấm Công — Time
Document #	2
Creator	ITBA Team
Date	17/06/2026
Version	2.0.0

1. EXECUTIVE SUMMARY

Time is a lightweight mobile attendance app (iOS & Android) designed for individual employees to self-track their daily working hours without requiring a company-side system. The app has no login or logout — it is a personal tool that runs persistently on the user's device.

The core loop: the app reminds the user to check in at a configured time, tracks NET working hours through a visual progress ring, handles a fixed lunch break (12:00–13:00), sends reminders throughout the day, and records attendance history in a monthly report. Users can also mark days as Leave. No authentication, no server-side user management.

2. PROJECT DESCRIPTION

Time covers 3 main screens: - **Home** — progress ring, check-in/check-out, shift selection, lunch break display - **Report** — monthly attendance calendar, day detail, import/export - **Profile & Settings** — avatar, name, attendance period, daily reminder time

The app operates entirely from the user's device. Check-in and check-out are self-reported. Working hours are calculated as NET (lunch excluded) for the ring/goal and GROSS (check-out minus check-in) for Report and Widget display.

3. PROJECT SCOPE

Estimated duration: ~3 months **Team:** 01 PM, 01 BA, 02 Mobile Dev, 01 BE, 01 Tester, 01 UI/UX - Elicitation & Design: 1 month - Development: 1.5 months - Testing & QA: 0.5 months

IN-SCOPE	OUT-OF-SCOPE
Profile setup (avatar, name, attendance period, reminder time)	Login / Logout / Authentication
Check In — Morning Shift (8h NET goal)	Admin dashboard
Check In — Afternoon Shift (4h NET goal, available after 12:00)	Company-level management
Leave Day marking	Payroll integration
Lunch break auto-pause (12:00–13:00, NET hours only)	Biometric / GPS check-in
Progress ring (NET hours real-time)	Team / department reporting

IN-SCOPE	OUT-OF-SCOPE
Widget — shows check-in time, check-out time, GROSS hours	Leave approval workflow
Monthly attendance report (calendar view)	Overtime compensation tracking
Day detail with edit capability	Multi-device sync
Summary card (on-time / incomplete / missing / leave counts)	
Import data (CSV)	
Export data	
Reminder system (daily, 1-hour warning, completion, 18:00 check-out prompt)	

4. BUSINESS DRIVERS

Driver	Description
BD1: Reduce missed attendance records	Automated reminders prevent forgotten check-ins/outs, keeping attendance data complete without HR intervention.
BD2: Individual accountability	Employees self-track their hours in real-time, taking personal responsibility for attendance without depending on a company system.
BD3: Flexible shift support	Morning (8h) and Afternoon (4h) shift options accommodate diverse work schedules within one app.
BD4: Fast-launch, learn early	Focused Phase 1 scope enables a 3-month launch to gather real user feedback before Phase 2 expansion.

5. PRESENT PROCESS

Details: Employees manually remember to record check-in/out via paper timesheets or a desktop-only HR portal. No reminders exist. Missed records require HR correction, delaying payroll.

Flow: 1. Employee arrives at work [employee] 2. Remembers/forgets to log check-in [employee] 3. Remembers/forgets to log check-out [employee] 4. Month-end: HR reviews records, finds errors [HR] 5. Employee submits correction request [employee] 6. HR manually approves and corrects [HR] 7. Corrected records used for payroll calculation [HR]

6. PROPOSED PROCESS

With Time app, the employee receives a reminder at the configured time. They open the app, select their shift, and tap Check In. The progress ring counts NET working hours. At 12:00 the ring pauses for lunch and resumes at 13:00. The system sends reminders as the shift nears completion and at 18:00 if check-out hasn't been recorded.

Full daily flow:

1. System sends daily reminder at user-configured time: "Have you checked in?" [system]
2. User opens Time app [user]

3. User selects shift: **Morning Shift** (8h goal) or **Afternoon Shift** (4h goal, available after 12:00) or **Leave Day** [user]

Morning Shift path: 4. User taps Check In → system records check-in timestamp [system] 5. Progress ring starts counting NET hours toward 8h goal [system] 6. At 12:00 → ring pauses, turns orange, displays "Lunch Time" [system] 7. At 13:00 → ring resumes automatically [system] 8. System sends reminder: "1 hour remaining to complete your shift" [system] *(7h NET reached)* 9. System sends reminder: "You've completed 8 hours — do you want to check out?" [system] *(8h NET reached)* 10. If user has not checked out by 18:00 → system sends reminder: "Do you want to check out?" [system] 11. User taps Check Out → system records check-out timestamp [system] 12. Report updated: GROSS hours = check-out time – check-in time (lunch included in GROSS) [system] 13. NET hours = GROSS – 1h lunch (if check-out is after 13:00) [system]

Afternoon Shift path: 4a. User selects Afternoon Shift before 12:00 → Check In button disabled, ring shows countdown to 12:00 [system] 4b. At/after 12:00 → Check In button becomes available [system] 4c. User taps Check In → 4h NET goal timer starts [system] 4d. System sends reminder: "1 hour remaining" at 3h NET mark, "Goal complete" at 4h NET [system]

Leave Day path: 4x. User selects Leave Day → system records Full Day Leave for today [system] 4y. All remaining reminders for today are cancelled [system] 4z. Leave recorded in Report calendar for the day [system]

---

7. FUNCTIONAL REQUIREMENTS

Priority Legend

Value	Status	Description
5	Immediate	Critical — project not possible without this
4	High	High priority; project can ship MVP without it
3	Moderate	Valuable but not MVP-blocking
2	Low	Nice to have

PROFILE & SETTINGS

ID	Requirement	Priority
FR-001	**Avatar Setup:** Default avatar is the app logo. User can upload a photo from the device library. Photo is auto-cropped to circle. Tapping Save replaces the logo with the selected photo everywhere in the app (Home header, Report). Save required — changes not persisted until Save is tapped.	3
FR-002	**Name Setup:** User can enter or edit their display name shown in the Home greeting ("Hello [Name]"). Save required — changes not persisted until Save is tapped. Default if no name set: display "Hello" without a name.	3
FR-003	**Attendance Period Setup:** User sets a monthly cutoff date (1–28). System calculates the attendance period as: cutoff day of previous month → (cutoff day – 1) of current month. Example: cutoff = 26 → period = 26th prev month to 25th current month. This period is used in the Report summary card. Save required.	4
FR-004	**Daily Check-In Reminder Time:** User sets the time for the daily morning reminder (default: 08:45). At this time each day, the system sends a push notification: "Have you	5

ID	Requirement	Priority
	checked in?" Notification is not sent if the user has already checked in that day. Save required.	

HOME SCREEN

ID	Requirement	Priority
FR-005	**Home Screen Layout:** Three zones: (1) Header — greeting "Hello [Name]", today's date (format: Weekday, DD Month), avatar (top right); (2) Hero — progress ring + hours display + shift info + CTA button; (3) Stats — 2 mini cards showing this month's on-time count and leave count, tap navigates to Report.	5
FR-006	**Check In — Morning Shift:** User taps Check In → selects Morning Shift (or default if no selection) → system records timestamp as check-in time → Check Out button appears → 8h NET goal timer starts. Check-in time is recorded in the Report card and Widget immediately. Cannot check in twice in the same day without checking out first.	5
FR-007	**Check In — Afternoon Shift:** User selects Afternoon Shift. If selected before 12:00 → Check In button is disabled; ring displays a countdown until 12:00. At 12:00, Check In becomes available. User taps Check In → system records timestamp → 4h NET goal timer starts. Lunch pause (12:00–13:00) does not apply if user checks in after 13:00.	4
FR-008	**Check In — Leave Day:** User taps Leave Day → system records Full Day Leave for today in the Report. No check-out required. Progress ring is not shown. All remaining daily reminders for today are cancelled after Leave Day is recorded.	4
FR-009	**Progress Ring:** SVG circle 140×140px. Displays NET hours worked as a percentage of the shift goal (8h Morning / 4h Afternoon). Fill = color-primary (green). Center display: "Xh Ym" (hours/minutes worked NET) with "of Xh goal" subtitle. Updates every minute. States: empty ring (0%) before check-in → grows as hours accumulate → ring stroke turns orange (color-warning) when < 30 minutes remaining → ring turns full green when goal is reached → ring turns orange and freezes 12:00–13:00 with "Lunch Time" label.	5
FR-010	**Lunch Break Auto-Pause (12:00–13:00):** At exactly 12:00, the progress ring pauses and turns orange (color-warning). The label "Lunch Time" is displayed inside or below the ring. The NET hour count stops accumulating. At exactly 13:00, the ring resumes its previous color and continues counting. This 1-hour window is fixed and not configurable in Phase 1. Applies to Morning Shift only. NET calculation: if check-out is after 13:00, subtract 1h from GROSS to get NET.	5
FR-011	**Check Out:** Check Out button is available after check-in. Tapping Check Out records the check-out timestamp. System calculates: GROSS hours = check-out time – check-in time. NET hours = GROSS – 1h if check-out is after 13:00 and shift is Morning. Report and Widget updated with GROSS hours. If user attempts to check out with less than 4h NET (Morning) or 2h NET (Afternoon), a confirm dialog appears: "You have only worked Xh — confirm check out?"	5
FR-012	**Report Card on Home:** After check-in, the Home screen displays a card showing: check-in time, current working status. After check-out: shows check-in time, check-out time,	4

ID	Requirement	Priority
	GROSS total hours. This card is always visible on Home once a check-in exists for the day.	

WIDGET

ID	Requirement	Priority
FR-013	**Home/Lock Screen Widget:** Displays today's check-in time, check-out time (if checked out), and GROSS total hours. Widget reflects GROSS hours (check-out – check-in, lunch NOT deducted) consistent with Report. Updates in real-time after any check-in or check-out event.	3

REMINDER SYSTEM

ID	Requirement	Priority
FR-014	**Daily Check-In Reminder:** At the time configured in Profile (FR-004), system sends push notification: "Have you checked in?" Fired every working day. Not sent if user has already checked in that day.	5
FR-015	**1 Hour Remaining Reminder:** When NET working hours reach (shift goal – 1h), system sends push notification: "1 hour remaining to complete your shift." Fires at 7h NET for Morning Shift, 3h NET for Afternoon Shift.	4
FR-016	**Shift Complete Reminder:** When NET working hours reach the shift goal (8h or 4h), system sends push notification: "You've completed your shift — do you want to check out?"	4
FR-017	**18:00 Check-Out Prompt:** If the user has checked in but not checked out by 18:00, system sends push notification at 18:00: "Do you want to check out?" Fires once regardless of shift type. Does not fire if already checked out or if Leave Day was selected.	4

REPORT SCREEN

ID	Requirement	Priority
FR-018	**Monthly Calendar View:** Displays a calendar grid (Mon–Sun, 7 columns) for the current attendance period (based on user-set cutoff date in FR-003). Each day cell shows a status dot (6px) below the date number. Navigation: month picker to move between periods; cannot navigate to a future period. Current day has a color-primary border.	5
FR-019	**Day Status Logic:** Each day is assigned one of four statuses based on recorded data for that day (all within the same calendar day — no overnight spanning): **On-Time** (green) = has both check-in AND check-out recorded; **Incomplete** (orange) = has check-in OR check-out, but not both; **Missing** (red) = no check-in AND no check-out AND not a Leave Day; **Leave** (blue) = marked as Full Day Leave via FR-008. Weekend days and public holidays: gray, not counted in summary.	5

ID	Requirement	Priority
FR-020	**Summary Card:** Displayed at the top or bottom of the Report screen. Shows total count of each status for the current attendance period: On-Time count / Incomplete count / Missing count / Leave count. Tapping a status filters the calendar to show only days with that status. Total working days = calendar days in period excluding weekends.	4
FR-021	**Day Detail Panel:** Tapping a day opens a bottom sheet showing: date, check-in time, check-out time, GROSS total hours, NET total hours, and day status. If no data: "No information for this day." If Leave Day: "Full Day Leave." Includes an Edit button (FR-022).	4
FR-022	**Edit Day Record:** From the Day Detail panel, user can tap Edit to manually correct: check-in time, check-out time, or day status (On-Time / Incomplete / Missing / Leave). Edited records are marked as manually edited. System recalculates GROSS and NET after edit. Save required to persist changes.	4
FR-023	**Import Data (CSV):** User can import attendance data from a CSV file. Default format: columns = Date, Check-In, Check-Out, Status. System maps imported data to the Report calendar. Conflicts (existing data vs. imported data) prompt user to choose: keep existing or overwrite.	3
FR-024	**Export Data:** User can export attendance records for the current attendance period as a CSV file. Export includes: Date, Check-In, Check-Out, GROSS Hours, NET Hours, Status.	3

8. NON-FUNCTIONAL REQUIREMENTS

ID	Requirement	Category
NF1	Check-in and check-out actions must complete and confirm within 2 seconds on a 4G connection.	Performance
NF2	Push notifications must be delivered within 30 seconds of the scheduled trigger time.	Performance
NF3	Monthly calendar must load within 3 seconds on a 4G connection.	Performance
NF4	Progress ring must update every 60 seconds minimum in real-time.	Performance
NF5	Lunch break pause must be handled entirely client-side — no server call required to pause or resume. Ring must resume exactly at 13:00 even if the app is backgrounded.	Performance
NF6	App must support iOS 14+ and Android 10+ (minimum 3 OS versions below the latest).	Compatibility
NF7	All interactive elements must have a minimum touch target of 44×44px.	Accessibility
NF8	Text color contrast must be ≥ 4.5:1 against all backgrounds (WCAG AA).	Accessibility
NF9	App must support Dynamic Type / system font scaling. Minimum font size: 11px.	Accessibility
NF10	Color must not be the sole status indicator — icon or text label must accompany all color-coded elements (dots, badges, ring states).	Accessibility

ID	Requirement	Category
NF11	All color values must use design token variables — no hardcoded hex values in source code.	Maintainability
NF12	Dark mode must be supported. All color tokens must have dark-mode variants. Both modes must be tested before release.	Usability
NF13	Attendance data must be persisted locally on the device to prevent data loss on crash or forced app close.	Data Integrity
NF14	GROSS and NET hours must be calculated consistently everywhere: GROSS = check-out – check-in; NET = GROSS – 1h if check-out is after 13:00 and shift is Morning. No discrepancy between Home, Report, and Widget values.	Data Integrity
NF15	Profile settings (avatar, name, attendance period, reminder time) must NOT be auto-saved — all changes require an explicit Save action to persist.	Data Integrity
NF16	All main features must be reachable within 2 taps from the Home screen.	Usability
NF17	Notification permission must be requested explicitly on first setup. Permission denial must be handled gracefully with a clear fallback message.	Security
NF18	Critical bugs (check-in failure, ring not updating, notification not firing) must be hotfixed within 24 hours.	Maintainability
NF19	No gradient backgrounds. No hardcoded colors. Minimum font size 11px. Touch targets ≥ 44×44px throughout.	Design Compliance

9. GLOSSARY

Term	Explanation
Morning Shift	Standard full-day attendance: 8h NET goal. Check-in available from app open. Subject to 12:00–13:00 lunch pause.
Afternoon Shift	Half-day attendance: 4h NET goal. Check-in button available only at/after 12:00. Lunch pause does not apply if check-in is after 13:00.
Leave Day	User marks the day as a full-day leave. No check-in/out required. Recorded in Report as "Leave." All day's reminders are cancelled.
NET Hours	Actual working time = GROSS – 1h lunch (if check-out after 13:00 on Morning Shift). Used for the progress ring and shift goal.
GROSS Hours	Total time from check-in to check-out including lunch. Used for Report, Widget, and Day Detail display.
Progress Ring	SVG circle 140×140px on the Home screen displaying % of NET hours worked vs. the shift goal. Pauses and turns orange during lunch (12:00–13:00).
Lunch Time	The fixed 12:00–13:00 window where the progress ring pauses, turns orange, and displays "Lunch Time." Not counted in NET hours.

Term	Explanation
Attendance Period	User-defined monthly reporting cycle based on a cutoff date. Example: cutoff = 26 → period = 26th prev month to 25th current month.
On-Time	Day status: both check-in AND check-out recorded for that day.
Incomplete	Day status: only check-in OR only check-out recorded — not both.
Missing	Day status: no check-in and no check-out, and no Leave Day marked.
Widget	Home screen or lock screen widget showing today's check-in time, check-out time, and GROSS hours.
Design Token	Named design values (colors, fonts, spacing) that are the single source of truth for the app's visual system.

10. REFERENCES

Name	Location
UI Annotation Document v1.0	ChamCong/UI_Annotation_Document_v1.0.docx
Design Token Document v1.0	ChamCong/Design_Token_Document_v1.0.docx
TimeApp Masterplan (local)	ChamCong/TimeApp_masterplan.xlsx
TimeApp GG Sheets	https://docs.google.com/spreadsheets/d/1CJ0qKjeET9z7mDx8XEWbyj8f5JcgJMvnlEJ47VqvbLY

11. APPENDIX

Screen summary:

Home Screen - Header: "Hello [Name]", date, avatar - Hero: Progress ring (NET hours, 8h or 4h goal), Xh Ym center display, "of Xh goal" subtitle - Ring states: empty (before check-in) → fills green (checked in) → orange stroke (< 30 min remaining) → orange + "Lunch Time" (12:00–13:00) → full green (goal complete) - CTA: Check In button → opens shift selection (Morning / Afternoon / Leave Day) - After check-in: Check Out button replaces Check In; report card shows check-in time - Stats: 2 mini cards — on-time count / leave count this month

Report Screen - Month picker (cannot navigate to future period) - Summary card: On-Time / Incomplete / Missing / Leave counts; tap to filter calendar - Calendar grid: Mon–Sun, color-coded dots (green/orange/red/blue/gray) - Day detail bottom sheet: check-in, check-out, GROSS, NET, status, Edit button - Import (CSV) and Export buttons

Profile & Settings Screen - Avatar (default: app logo; changeable via photo library, crops to circle) - Name (editable text field) - Attendance Period (cutoff date picker) - Daily Check-In Reminder time picker (default 08:45) - Single Save button — all 4 settings persist only on Save

Reminder Schedule (per day): 1. [User-configured time] → "Have you checked in?" (skipped if already checked in) 2. [Check-in time + shift goal – 1h NET] → "1 hour remaining" 3. [Check-in time + shift goal NET] → "You've completed your shift — check out?" 4. [18:00] → "Do you want to check out?" (only if checked in, not yet checked out)

Hours calculation rule: - GROSS = check-out – check-in (always, including lunch) - NET = GROSS – 1h IF check-out is after 13:00 AND shift is Morning - NET = GROSS IF check-out is at or before 13:00 OR shift is Afternoon - Report, Widget, Day Detail always show GROSS - Progress ring and shift goal always use NET